Preventive Maintenance of Machining Centre MC-500

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Effective date	2026-01-20	**Owner**	Maintenance
Classification	Internal		

1 Scope

This manual lists the preventive maintenance tasks for the machining centres MC-500 (asset numbers MC-501 to MC-506). Before any work inside the machine, the machine must be locked out according to MP-LOTO-002.

2 Maintenance intervals

Task	Interval	Responsible
Clean chip conveyor and coolant filter	weekly	operator
Check way lube oil level	weekly	operator
Check spindle drawbar clamping force	every 500 operating hours	maintenance
Change hydraulic oil (HLP 46, 40 l)	every 4000 operating hours	maintenance
Clean and inspect axis covers	every 1000 operating hours	maintenance
Geometry check (ballbar test)	every 6 months	maintenance
Replace coolant completely and clean tank	every 3 months	maintenance

3 Spindle drawbar clamping force

The nominal clamping force of the HSK-A63 drawbar is 18 kN. Measure it with the clamping force gauge. If the force is below 15 kN, the disc springs must be replaced. Do not run the machine with a clamping force below 15 kN, the tool can be pulled out during heavy cuts.

4 Spindle warm-up

After a standstill of more than 8 hours, run warm-up program O9001. The program runs the spindle for 10 minutes in steps of 2000, 6000 and 10000 rpm. Without warm-up, thermal growth of the spindle can cause dimensional errors of up to 0.015 mm in the first parts.

5 Ballbar test

Run the ballbar test in the XY plane with radius 150 mm at feed 1000 mm/min. The circularity must be below 12 um. If it is higher, inform the machine supplier service.

6 Spare parts

Part	Order number	Minimum stock
Disc spring set drawbar	MC5-SP-118	2
Coolant filter cartridge	MC5-FL-020	6
Way lube oil 5 l	LUB-068-5	4

7 Revision history

Rev	Date	Change
A	2024-11-04	First issue
B	2026-01-20	Coolant change interval 6 -> 3 months